LAB-III TASK 4 lab Report

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Section B

Title:

Implementation of If-Else Statement in C

Objective:

To use the `if-else` statement to determine whether a person is eligible to vote based on age.

Theory:

The `if-else` statement is used when there are two possible outcomes. It allows the program to execute one block of code if a condition is true and another block if the condition is false.

Syntax:

```
if (condition)
{
    // executed if condition is true
}
else
{
    // executed if condition is false
}
```

In this task, we check whether a person’s age is greater than or equal to 18.

Program (with comments):

```
#include <stdio.h>   // Input-output library

int main()
{
    int age; // Declare variable to store age

    printf("Enter your age: "); // Prompt user
    scanf("%d", &age);          // Read input

    if (age >= 18)              // Check if age is 18 or above
    {
        printf("You are eligible to vote.\n"); // True condition
    }
    else                        // If condition is false
    {
        printf("You are NOT eligible to vote.\n"); // False condition
    }

    return 0; // End program
}
```

Detailed Explanation:

The program starts by declaring an integer variable `age` to store the user's input.

The user is prompted to enter their age, which is stored using `scanf`.

The condition `age >= 18` is evaluated:

- If true, the program prints that the user is eligible to vote.
- If false, the `else` block executes and prints that the user is not eligible.

This structure ensures that **one and only one block executes**, making it useful for binary decisions.

Output (Example):

Enter your age: 16
You are NOT eligible to vote.

Conclusion:

The `if-else` statement is useful for handling situations with two possible outcomes, improving decision-making in programs.